Analyzing the Relationships Between Depression, Test Anxiety, Self-Esteem, and Academic Achievement in Adolescents: A Mediating Effect Analysis

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INTRODUCTION

- Adolescence is a period of rapid physical, psychological , and social changes.
- In South Korea, academic achievement is considered a crucial developmental task that significantly influences self-esteem, career preparation, and quality of life.[1]
- **Depression and Test Anxiety are major factors that hinder academic success by reducing coping abilities and social engagement.[2]**
- Therefore, managing mental health during adolescence is essential to promote academic achievement and overall development.

OBJECTIVES

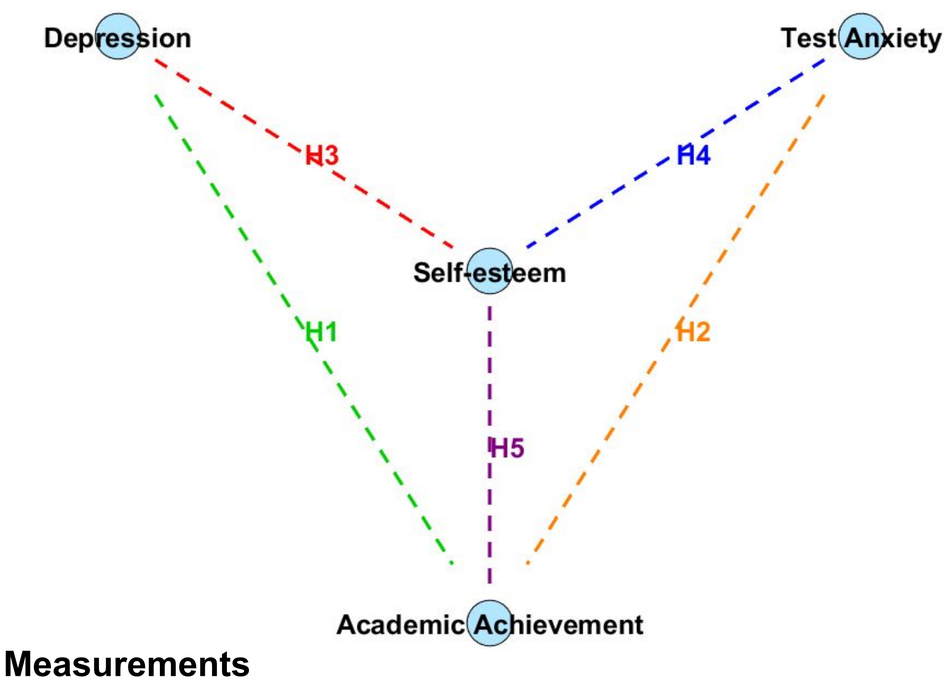
- To analyze the impact of depression and anxiety on academic achievement in adolescents.
- To provide insights for improving mental health and academic outcomes in educational settings.

METHODS

Participants

- A total of 60 participants (male: 35, female: 25)
- Among the participants, 46 high school students (Age range ≥16 years, male: 25, female: 21) were selected for analysis.

Study Design and Hypotheses



Measurements

- **Depression[3]:** Assessed using the CES-D, a 20-item scale measuring depressive symptoms across four dimensions. Higher scores indicate greater severity of depressive symptoms.
- **Test Anxiety[4,5]:** Measured with the K-TAI-K, a 20-item scale assessing cognitive and emotional test anxiety. Higher scores reflect higher levels of test anxiety.
- **Self-Esteem[6]:** Evaluated using the RSES, a 10-item scale measuring positive and negative self-perceptions. Higher scores represent higher self-esteem.

Reference

1. Park, S. (2022). Examining trajectories of early adolescents' life satisfaction in South Korea using a growth mixture model. *Applied Research in Quality of Life*, *17*(1), 149-168.
2. 신은희. (2020). 간호대학생의 학업성취도와 우울, 불안및 그릿과의 관계. *한국학교보건학회지 제*, *33*(3).
3. Radloff, L. S. (1977). The CES-D scale: A self-report depression scale for research in the general population. *Applied Psychological Measurement*, 1(3), 385–401.
4. Spielberger, C. D. (1980). *Test Anxiety Inventory: Preliminary professional manual*. Consulting Psychologists Press.
5. 김문주. (1991). 우리나라 학생들의 시험불안 진단을 위한 도구 개발 과정. *소아청소년정신의학*, *2*(1), 32-42.
6. Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton University Press.

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RESULTS

Analysis of Factors Influencing Depression in Adolescents

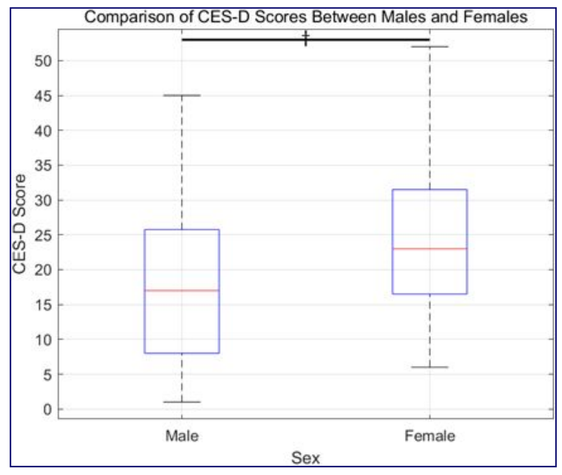


Fig 1. No significant difference in CES-D scores between sex

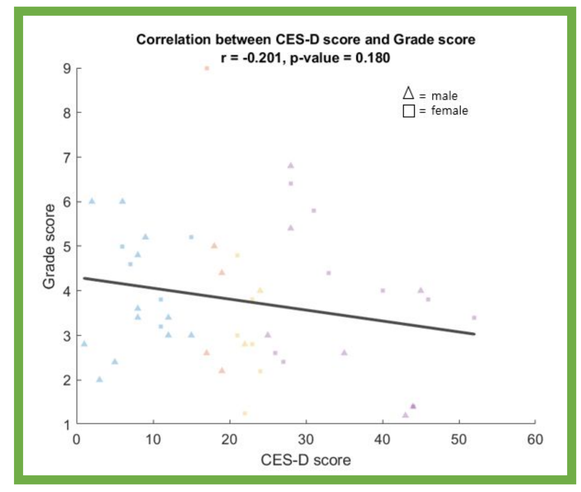


Fig 2. No significant correlation between depression and academic achievement

Analysis of Factors Influencing Test Anxiety in Adolescents

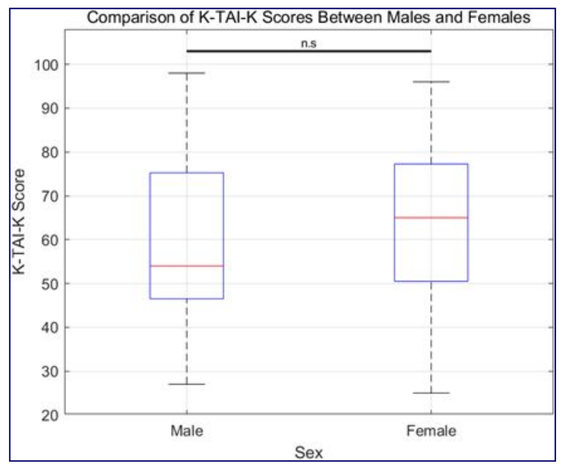


Fig 3. No significant difference in K-TAI-K scores between sex

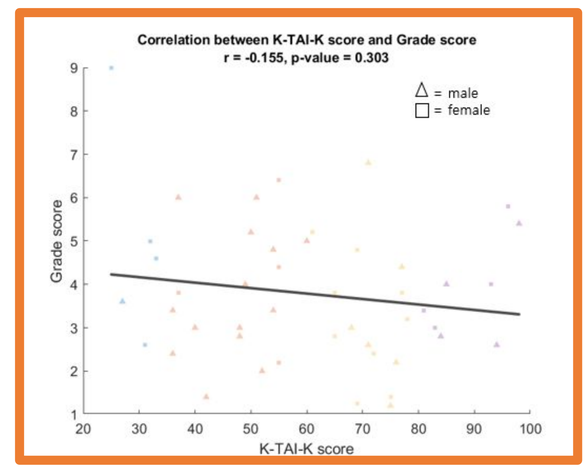


Fig 4. No significant correlation between test anxiety and academic achievement

Analysis of Factors Influencing Self-esteem in Adolescents

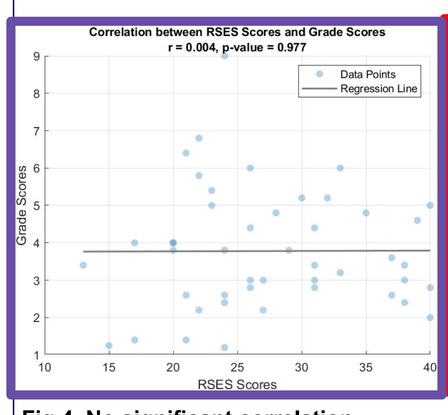


Fig 4. No significant correlation between self-esteem and academic achievement.

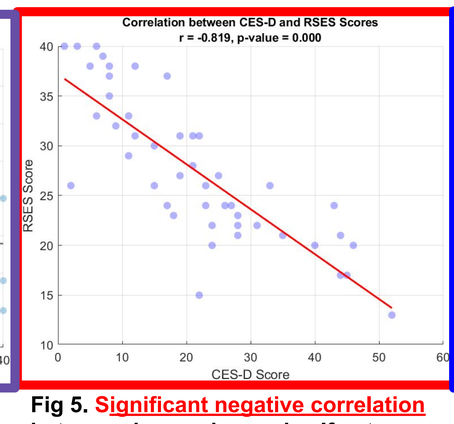


Fig 5. Significant negative correlation between depression and self-esteem

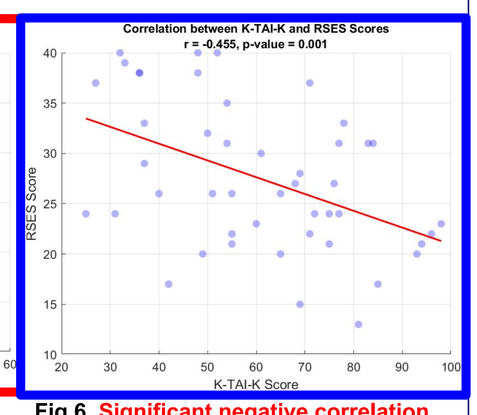
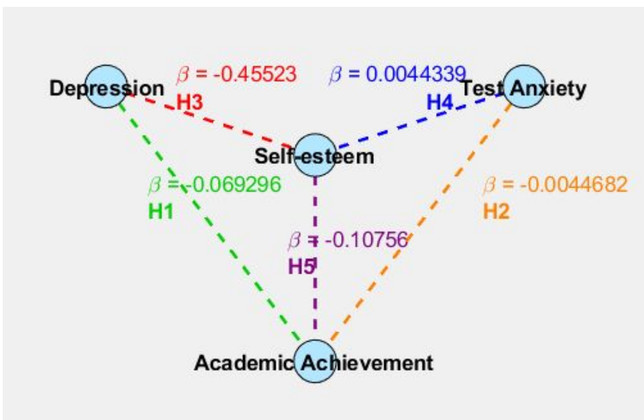


Fig 6. Significant negative correlation between test anxiety and self-esteem



Mediation Analysis Results

- **Self-esteem (RSES) has a moderate negative impact on academic achievement** (β = -0.107, significant).
- **Depression (CES-D) has a strong negative impact on self-esteem** (β = -0.455, significant).

****p<.0001, ***p<.001, **p<.01, *p<.05, †p<.10, n.s p>.10
β<.10 weak, .10≤β<.30 moderate, β≥.30 strong effect

CONCLUSIONS

- Depression negatively impacts self-esteem and indirectly lowers academic achievement.
- Test anxiety negatively affects self-esteem but not academic achievement indirectly.
- Enhancing self-esteem is key to mitigating depression's effects on academic achievement. So, self-esteem-focused interventions are essential.
- Ultimately, Managing depression and test anxiety supports both mental health and academic success.

연계전공 뇌–마음–행동